

The World Might Not Be as Complex as We Thought

The Roles of Scale, Perspectives, and Information in Human Understanding

Honggao Cao
September 2026

Abstract

The world often appears complex because the relationships available to an analyst are incomplete, fragmented, differently expressed across scales, or differently projected across perspectives. This paper develops a theoretical framework for analyzing that apparent complexity. The framework distinguishes two stages in the formation of human understanding. First, a world, object, or process becomes available as information under a particular scale and perspective. Second, an analyst organizes that information into an understanding of the phenomenon. Apparent complexity can arise in either stage: a relevant relationship may not be revealed under the operative conditions, or relevant informational elements may be available while the relations among them remain unrecognized. The framework is developed through physical, temporal, organizational, and representational examples, and then extended to cases in which individually intelligible descriptions obtained under different conditions are themselves difficult to connect. Historical work on perspectival observation, levels and scale, intertheoretic connection, information, and distributed cognition is brought into the discussion where it helps clarify particular elements of the framework. The resulting account does not treat complexity as a single property to be measured. It provides a way to ask why a phenomenon appears complex to an analyst and where that apparent complexity may enter the process of revelation and understanding.

Keywords: apparent complexity; scale; perspective; information; human understanding; relational fragmentation; cross-scale relations

1. Apparent Complexity

A three-dimensional cuboid does not change when an observer moves around it, yet its visible shape changes with the viewing angle. Three hydrogen atoms have mass, yet an ordinary household scale cannot establish a useful relationship between atom count and measured weight at that range. A reader can understand every sentence in a paragraph and still fail to see the argument until the relations among the sentences become clear. These examples differ in substance, but each presents the same broad possibility: what appears complicated, disconnected, or difficult to understand can depend on how a phenomenon becomes available to an observer and how the resulting information is organized.

This paper uses apparent complexity as a term for an analyst's encounter with a phenomenon as fragmented, irregular, disconnected, or discontinuous. The term describes the form in which the phenomenon becomes intelligible to the analyst; it does not by itself characterize the intrinsic structure of the world. Apparent complexity is therefore not the same as error. A perspective-dependent projection can be accurate, a local description can be useful, and a macro-level relation can legitimately differ in form from a micro-level

relation. The question is why these forms of difficulty arise and when a more connected account of the same phenomenon may become available.

The paper develops a theoretical framework for explaining how such apparent complexity can arise. It distinguishes between the conditions under which the world is revealed as information and the process by which that information becomes organized into understanding. This distinction allows apparent complexity to be analyzed in two places: first, when a relevant relationship is not revealed under the operative scale or perspective; and second, when the relevant informational elements are available but remain insufficiently connected in understanding.

The discussion proceeds from this basic structure. Section 2 introduces the framework and notation, explaining how scale and perspective condition what becomes available as information and how an analyst transforms that information into an understanding of the phenomenon. Section 3 examines conditioned revelation: cases in which a relationship may be present in the world but absent from the information available under a particular scale or perspective. Section 4 turns to relational fragmentation, where informational elements may already be available but the relationships among them remain unrecognized; the section also explains how connected understanding differs from merely having more informational elements.

Section 5 develops scale and perspective more fully. Scale concerns the level, range, duration, or aggregation at which a phenomenon is encountered; perspective concerns the position or orientation from which that accessible phenomenon is observed. The examples in this section show that changing either condition can reveal not only additional information but different relationships. Section 6 then considers a further problem created by those differences: two descriptions may each be intelligible under their own conditions while the relationship between them remains unclear. The section develops this connection problem and considers how changing the conditions of inquiry can sometimes help expose the missing relation. Section 7 brings historical and neighboring work into the framework where it helps readers connect the paper's elements to earlier intellectual traditions. Section 8 draws the argument together in discussion and conclusion.

2. A Framework of World Revelation and Human Understanding

2.1 From the world to information

Consider a world, object, or process of interest, W . We encounter W under some scale, denoted by σ , and from some perspective, denoted by π . Scale describes the level or range at which W is accessed. Depending on the problem, it can refer to physical size, duration, measurement range, or level of aggregation.

Perspective describes the position or orientation from which the accessible phenomenon is encountered.

The notation below gives the framework a compact and consistent language. Its purpose is to make the two stages, their relationships, and the locations at which apparent complexity can arise easy to identify and refer to. The notation organizes the conceptual argument; it is not intended as a quantitative predictive model.

Let I denote the information about W that becomes available under those conditions. We can write:

$$I = \mathcal{R}(W; \sigma, \pi) \tag{1}$$

Equation (1) says that the available information I is a revelation of W conditioned by scale σ and perspective π . The calligraphic $\mathcal{R}$ in Equation (1) names that revelation process. It does not require us to assume one

literal measurement function for every application. Its role is to preserve a simple point: an analyst does not receive the world without conditions.

A household scale and a molecular measurement regime illustrate a difference in scale. Two observers looking at the same cuboid from different angles illustrate a difference in perspective. Both differences can change what information becomes available even though the underlying object has not been replaced by another object.

2.2 From information to understanding

Information is not yet understanding. An analyst may possess many observations, measurements, records, or statements without recognizing the relation that makes them jointly intelligible. Let A denote the analyst and U the understanding formed from the available information. We can write:

$$U = \mathcal{G}_A(I) \quad (2)$$

Equation (2) says that understanding U results from an analytical operation applied to the available information I . The calligraphic G in Equation (2) emphasizes that information must be organized before it becomes an intelligible account. The same informational material can support different levels of understanding, and relevant relationships can remain unrecognized even when the underlying informational elements are already available.

Putting the two stages together gives the framework used throughout the paper:

$$U = \mathcal{G}_A(\mathcal{R}(W; \sigma, \pi)) \quad (3)$$

The key elements and notations used in the framework are summarized in Table 1.

Table 1. Key Elements and Notation of the Framework

Symbol	Meaning	Plain-language reading
W	World, object, or process of interest	What is being encountered or studied
σ	Scale	At what level, range, duration, or aggregation W is accessed
π	Perspective	From what position or orientation W is encountered
I	Information	What becomes available under the operative scale and perspective
A	Analyst	The person or analytical process organizing the information
U	Understanding	The account formed from the available information
$\mathcal{R}$	Revelation mapping	How W becomes available as information under σ and π
$\mathcal{G}_A$	Understanding mapping	How analyst A organizes information into U

3. Conditioned Revelation

The first source of apparent complexity lies between W and I . A relationship can belong to the phenomenon under study yet fail to appear in the information revealed under a particular scale or perspective.

Let r denote a relationship of interest. A case of non-revelation can be represented as:

$$r \in W, \quad r \notin I \quad (4)$$

Equation (4) says that the relationship belongs to the phenomenon but is not present in the information available to the analyst under the operative conditions. The immediate implication is:

$$r \notin I \not\Rightarrow r \notin W \quad (5)$$

Failure to observe a relationship therefore does not establish that the relationship is absent from the world.

The hydrogen-atom example makes the point with little conceptual machinery. Total mass changes with atom count, but an ordinary household scale does not resolve differences at the level of a few atoms. Repeating the measurement can increase the number of observations without changing the kind of information the instrument can reveal.

This example exposes a distinction that matters beyond measurement. Sometimes a relationship is available under the current conditions but is masked by noise, sparse observations, or limited precision; in that case, more information of the same kind can help. In other cases, the operative scale or perspective does not make the relationship available in the required form at all. The difficulty is then not insufficient quantity but condition-limited revelation. No amount of repetition under unchanged conditions supplies information that those conditions do not reveal.

More information of the same kind is not necessarily the same as different information revealed under different conditions.

Conditioned revelation is one reason apparent complexity can arise. An analyst may confront an incomplete or fragmented picture not because the world lacks structure, but because the operative conditions do not expose the relevant relationship. Section 5 develops the stronger case further: some relational objects, such as variance, become meaningful only at a different scale.

4. Relational Fragmentation and Connected Understanding

Apparent complexity can also persist after the relevant information has reached the analyst. The problem is then not primarily one of non-revelation, but of organization.

Suppose the information available to the analyst consists of identifiable informational elements:

$$I = \{i_1, i_2, \dots, i_n\} \quad (6)$$

An informational element i_j can be an observation, measurement, statement, record, or other item of information relevant to the problem. Equation (6) says only that these elements are available. It does not say that the relations among them are understood.

Having the informational elements available does not by itself make their relationships available to the analyst. Those relationships can involve sequence, dependence, comparison, qualification, aggregation, or other forms of connection relevant to the problem.

A collection can therefore be complete as a list of informational elements while remaining incomplete as an intelligible structure.

This leads to a second implication:

$$r \notin U \not\Rightarrow r \notin I \quad (7)$$

Equation (7) says that failure to understand a relationship does not establish that the informational basis for that relationship is absent. The relevant relation may already be recoverable from available elements even though it has not yet been incorporated into the analyst's understanding.

Language provides an intuitive case. A reader can understand the meaning of each sentence in a paragraph while missing how one sentence qualifies another, how several sentences jointly support a claim, or how a conclusion follows from what came before. The difficulty is not missing sentences. It is missing relational organization.

Organizational behavior provides another case. Individual actions can each make sense from a local position while the organization as a whole appears inconsistent. Broader information about authority, dependencies, incentives, or sequencing can reveal relations among those actions and make the joint pattern more intelligible.

Connected understanding therefore requires more than accumulation. It requires the analyst to recognize and organize the relationships among the informational elements that matter for the problem.

The distinction is important: the analyst may already possess the relevant informational elements while still lacking the relational organization needed for a connected account.

This helps explain why apparent complexity can grow even as knowledge grows. Additional reports, variables, measurements, or specialist descriptions can enlarge the analyst's inventory while leaving the cross-relations unresolved. The result can be more information and more apparent complexity at the same time.

5. Scale and Perspective in Apparent Complexity

The preceding sections identify two places where apparent complexity can enter the framework. This section examines more closely the two conditions that shape revelation in the first stage: scale and perspective. They are analytically distinct but can interact in practice. Scale concerns the level or range at which a phenomenon is accessible; perspective concerns the position or orientation from which that accessible phenomenon is encountered.

5.1 Scale: different levels can reveal different relationships

Scale can affect what is detectable, but its role is broader than measurement resolution. A change of scale can change the relational description that is useful or even meaningful.

Consider individual observations that satisfy:

$$y_i = x_i^2 \quad (8)$$

At the level of individual observations, Equation (8) gives a simple relationship between x_i and y_i . At a population level, taking expectations gives:

$$\mathbb{E}[y] = \mathbb{E}[x^2] = (\mathbb{E}[x])^2 + \text{Var}(x) \quad (9)$$

In plain language, the average of the squared individual values depends not only on the square of their average but also on how much the individual values vary. Variance is therefore a population-level quantity defined across observations rather than another individual value. The example shows that a relational object can become meaningful at one scale even though it is not part of the natural description at another.

Temporal scale works similarly. An analyst can observe a state at a particular moment:

$$S_t = [x(t), y(t)] \quad (10)$$

Yet the relationship of interest may unfold across a broader process:

$$x(t_0) \rightarrow z(t_1) \rightarrow y(t_2) \quad (11)$$

Here $z(t_1)$ represents an intermediate step in the process that is not visible in a single-time snapshot. A narrow temporal scale can therefore expose states while hiding the process connecting them. This is why causal attribution is relevant here: a problem that looks difficult at one temporal scale can become differently structured when observation is widened to include the intervening process.

Traffic offers a less formal example. Local braking and merging can look irregular to an individual driver, while network-level information about bottlenecks, merge points, and flow propagation can make parts of that irregularity intelligible. Local unpredictability and higher-level intelligibility can coexist.

5.2 Perspective: different vantage points can reveal different projections

Perspective concerns a different kind of conditioning. Hold the accessible object roughly fixed and change the observer's position relative to it. The cuboid provides the simplest case. A three-dimensional object with dimensions x , y , and z can present different visible shapes from different viewing angles even though the object itself has not changed.

If P_π denotes the projection associated with perspective π , then:

$$I_\pi = P_\pi(W) \quad (12)$$

Equation (12) says that the information revealed from one perspective is a projection of the same underlying W . Different perspectives can therefore yield different but compatible descriptions.

The idea extends beyond physical viewing angle. In an organization, position can determine which interactions, constraints, and dependencies are visible. An employee, a manager, and an external observer may encounter different informational projections of the same institutional process. The differences need not arise from disagreement or error; different vantage points can make different aspects and relationships of the same phenomenon available to the observer.

Scale and perspective can also change together. Ordinary terrestrial observation, for example, combines a local scale with a particular physical vantage. Broader astronomical access changes both what range of the system becomes observable and the vantage from which relationships can be reconstructed. The concepts remain distinct, but inquiry often alters them jointly.

The common result is that revelation can change qualitatively when scale or perspective changes. A different condition can expose a different relation, not merely add another observation to the existing set.

6. Across Conditions: Connection, Transition, and Apparent Discontinuity

Once scale and perspective can yield different but individually intelligible descriptions, a new problem appears. Understanding each description separately does not yet explain how the descriptions are related.

Suppose one set of conditions makes information I_1 available and another makes information I_2 available. Each information set can support an intelligible account under its own conditions:

$$I_1 = \mathcal{R}(W; \sigma_1, \pi_1), \quad I_2 = \mathcal{R}(W; \sigma_2, \pi_2) \quad (13)$$

Equation (13) represents information revealed from the same world or process under two different sets of conditions. From I_1 and I_2 , the analyst may form two individually intelligible understandings, U_1 and U_2 . The remaining question is how those understandings are related.

This creates a connection problem. Understanding U_1 and U_2 separately does not necessarily provide the relation that makes them jointly intelligible. We use the term bridge for whatever knowledge supplies that connection:

$$U_1 \overset{B_{12}}{\leftrightarrow} U_2 \quad (14)$$

In Equation (14), B_{12} denotes knowledge of the relation that connects the two conditioned understandings. A bridge is informative when it does more than assert that U_1 and U_2 are related: it explains why they differ and how both can belong to the same broader phenomenon. Depending on the problem, that connection can take the form of aggregation, a change of representation, a transition process, a limiting relation, or another relation that makes the cross-condition difference intelligible.

The scale example in Section 5 provides a simple bridge already available within the paper. At the individual level, $y_i = x_i^2$. At the population level, the corresponding average relation contains both the square of the mean and the variance. The identity $E[x^2] = (E[x])^2 + \text{Var}(x)$ connects the two descriptions by showing exactly what appears when individual observations are organized at population scale. It is a bridge because it explains the difference between the descriptions rather than merely placing them side by side.

The framework uses apparent discontinuity for the analyst's experience of two conditioned understandings as disconnected because the relation between them is not yet intelligible. U_1 and U_2 can each be coherent while B_{12} remains unknown. In such a case, the observed gap does not by itself establish a break in the underlying world; it identifies a connection that has not yet been explained.

Knowledge of two descriptions is not necessarily knowledge of the relation connecting them.

The connection problem also clarifies one role of inquiry. When a relation remains obscure under one set of conditions, inquiry need not consist only of collecting more information under those same conditions. The analyst can change scale or perspective and compare what becomes available:

$$(\sigma_1, \pi_1) \mapsto I_1, \quad (\sigma_2, \pi_2) \mapsto I_2 \quad (15)$$

Equation (15) represents a simple extension of the framework. Comparing information revealed under different conditions can itself provide evidence about what varies, what remains stable, and how two descriptions may be connected. This is the same reason that rotating the cuboid can reveal depth that no amount of repeated viewing from a fixed angle would provide.

The point is not that changing conditions always resolves apparent complexity. It is that the framework identifies a different route to understanding: seek different information by changing the conditions of revelation, and seek the relation that organizes descriptions obtained under those conditions.

7. The Framework in Relation to Prior Work

The framework developed here draws on ideas with substantial historical roots. Those antecedents do not supply the argument as a ready-made theory of apparent complexity, but they help clarify several of its elements and give readers intellectual routes into the formulation.

The first transition in the framework, from W to I, has clear connections to earlier work on observation and representation. Hanson argued that observation is shaped by conceptual commitments, and Kuhn emphasized that scientific perception is not independent of the paradigms within which inquiry proceeds [1,2]. Giere later formulated a scientific perspectivism in which models, instruments, and observational positions selectively represent features of the world [3], while Massimi's perspectival realism preserves the possibility of genuinely world-directed knowledge [4]. The present framework takes from this tradition the idea that access to the world is conditioned rather than transparent. It uses that idea more narrowly than a general thesis about scientific knowledge: the question here is how conditioned access can make a phenomenon appear more fragmented or difficult to an analyst.

The role assigned to scale also has recognizable predecessors. Oppenheim and Putnam organized scientific description into levels, Simon treated complexity through hierarchical organization, and Anderson's 'more is different' argument emphasized that higher-level behavior can require concepts not obvious from lower-level description [5-7]. Renormalization and later philosophical work on emergence and effective description provide more formal examples of how useful relations can change with scale [8-10]. The present framework takes from this work the possibility that scale can change the form of an intelligible relation, not merely its resolution. It leaves open the stronger questions of whether the resulting description is emergent, reducible, or ontologically distinct; those questions are not needed for the analysis of apparent complexity.

Perspective has an equally direct historical connection. The present paper begins with the literal notion of vantage, illustrated by a cuboid, while scientific perspectivism provides a broader setting for the claim that what becomes available depends partly on the position and means through which a phenomenon is encountered [3,4]. The framework takes the selective-access insight but treats perspective as one conditioning variable within a larger structure, alongside scale. It therefore leaves aside the broader question of whether all knowledge should be characterized as perspectival.

The second transition, from I to U, connects to work on information, representation, and distributed cognition. Shannon's mathematical theory established a precise quantitative concept of information for communication systems [11], but the present framework uses information more broadly to denote what becomes available to an analyst under given conditions. Hutchins showed how cognitive accomplishment can depend on the organization and distribution of information across people, representations, and artifacts [12], while Bechtel and Richardson, Craver, and Wimsatt emphasize decomposition, organization, and relations among parts in scientific explanation [13-15]. The framework takes from these traditions the

importance of organization and relation, while leaving aside any claim that human understanding can be reduced to information quantity or to one cognitive mechanism.

The bridge introduced in Section 6 also has a historical lineage. Nagel's treatment of intertheoretic reduction and Schaffner's later work asked how descriptions expressed in different theoretical vocabularies could be connected [16,17]. The present framework takes from that tradition the connection problem itself: possessing descriptions on both sides of a divide is not the same as knowing their relation. It broadens the idea of a bridge beyond deductive or reductive relations and leaves aside the requirement that one description must be reduced to the other.

The framework therefore sits near several established traditions without being reducible to any one of them. Its organizing target is apparent complexity. The historical connections are useful because they show that individual components of the framework have been encountered in other forms, while the present paper brings them together around the two-stage movement from world revelation to human understanding.

8. Discussion and Conclusions

The framework developed in this paper begins with a simple analytical separation. A phenomenon first becomes available as information under a scale and perspective; that information then has to be organized into understanding. Once those stages are distinguished, several forms of apparent complexity become easier to locate.

Some apparent complexity arises because a relevant relationship is not revealed under the operative conditions. A measurement scale can be too coarse, a temporal scale can be too narrow, or a perspective can project away an important dimension. In such cases, accumulating more observations under unchanged conditions may not provide the missing relation. Different information may require different conditions of revelation.

Other apparent complexity arises after the informational elements are already available. Facts, measurements, statements, or local explanations can remain disconnected because the relations needed to organize them are not represented in understanding. Relational fragmentation therefore explains why greater informational quantity can coexist with greater apparent complexity.

A further difficulty appears when different scales or perspectives yield understandings that are each intelligible but not yet connected. The bridge formulation isolates that problem. Knowing U_1 and U_2 does not necessarily provide B_{12} . Apparent discontinuity can therefore arise from missing connection knowledge even when the accounts on both sides are mature.

These observations suggest a simple diagnostic sequence for inquiry. When apparent complexity persists under fixed conditions, the analyst can ask whether a different scale or perspective would reveal a different kind of information. When information is abundant but remains incoherent, the problem may instead be relational organization. When two accounts are individually intelligible but jointly puzzling, the relevant task is to seek the relation that connects them. These questions do not guarantee simplification, and broader scale or wider perspective is not always better; a new condition can reveal additional distinctions as well as new connections. The value lies in locating the source of the apparent difficulty more precisely.

The paper's title follows from that possibility. The world can remain rich, multidimensional, and genuinely difficult while some of what appears complex reflects the conditions under which relationships become visible and connected. What looks like disorder at one scale can participate in a more intelligible pattern at another. What looks like contradiction from two perspectives can become compatible once their projections are understood. What looks like a collection of separate facts can become a connected account when their relations are recognized. What looks like discontinuity can sometimes be a missing bridge.

Human understanding therefore advances through more than accumulation. It can advance by changing the scale or perspective through which a phenomenon is encountered, by organizing informational elements relationally, and by discovering connections among descriptions produced under different conditions. In that sense, the world might not be as complex as we thought: not because complexity must disappear, but because some apparent complexity may belong to the path through which the world becomes intelligible to us.

AI-Use Disclosure

AI tools were used in the development of this working paper for exploratory discussion, literature discovery, drafting assistance, and editorial revision. The author selected the research question, developed and directed the argument, evaluated the resulting claims and structure, and takes responsibility for the final content.

References

- [1] Hanson, N. R. (1958). *Patterns of Discovery*. Cambridge University Press.
- [2] Kuhn, T. S. (1962). *The Structure of Scientific Revolutions*. University of Chicago Press.
- [3] Giere, R. N. (2006). *Scientific Perspectivism*. University of Chicago Press.
- [4] Massimi, M. (2022). *Perspectival Realism*. Oxford University Press.
- [5] Oppenheim, P., & Putnam, H. (1958). Unity of science as a working hypothesis. *Minnesota Studies in the Philosophy of Science*, 2, 3-36.
- [6] Simon, H. A. (1962). The architecture of complexity. *Proceedings of the American Philosophical Society*, 106(6), 467-482.
- [7] Anderson, P. W. (1972). More is different. *Science*, 177(4047), 393-396.
- [8] Wilson, K. G. (1971). Renormalization group and critical phenomena. I. Renormalization group and the Kadanoff scaling picture. *Physical Review B*, 4(9), 3174-3183.
- [9] Batterman, R. W. (2002). *The Devil in the Details: Asymptotic Reasoning in Explanation, Reduction, and Emergence*. Oxford University Press.
- [10] Noid, W. G. (2013). Perspective: Coarse-grained models for biomolecular systems. *Journal of Chemical Physics*, 139, 090901.
- [11] Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27, 379-423, 623-656.
- [12] Hutchins, E. (1995). *Cognition in the Wild*. MIT Press.
- [13] Bechtel, W., & Richardson, R. C. (1993). *Discovering Complexity: Decomposition and Localization as Strategies in Scientific Research*. Princeton University Press.
- [14] Craver, C. F. (2007). *Explaining the Brain: Mechanisms and the Mosaic Unity of Neuroscience*. Oxford University Press.

- [15] Wimsatt, W. C. (2007). *Re-Engineering Philosophy for Limited Beings: Piecewise Approximations to Reality*. Harvard University Press.
- [16] Nagel, E. (1961). *The Structure of Science*. Harcourt, Brace & World.
- [17] Schaffner, K. F. (1967). Approaches to reduction. *Philosophy of Science*, 34(2), 137-147.